

Yijing Wu

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EDUCATION

Northeastern University | Seattle, WA, U.S.

September 2021 – present

Master of Science in Information Systems; GPA: 3.9/4.0

- Courses: Web Design and User Experience Engineering, Smartphones-Based Web Development, Operating Systems

Tongji University | Shanghai, China

September 2017 – June 2021

Bachelor of Engineering in Communications Engineering

- Courses: Algorithms Design and Analysis, Principles of Database Systems, Computer Communication Networks, Software Tech
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SKILLS

Programming Languages: JavaScript, TypeScript, Python, JAVA, Swift, HTML, CSS, SQL, C/C++

Tools/Frameworks: React, React Native, Node.js, Django, MySQL, MongoDB, Docker, Xcode, Vue.js, AWS, GCP, Firebase, Git

EXPERIENCE

Upduo

San Francisco, CA

Software Engineer Co-op

June 2022 – December 2022

- Performed as a full-stack engineer in an agile team to build, improve and maintain cross-platform web&mobile applications
- Collaborated with product and design team and conducted **TypeScript** programming (**React** and **React Native** with **Redux** state container) to deliver new UI/UX for front-end, including graph data visualization, animations, kudos interface, etc.
- Worked on **Django** backend by **Python**, applied celery to manage asynchronous tasks, updated **GraphQL** while maintaining minimal calls to the **PostgreSQL** database, implemented unit and integration tests to validate the behavior
- Integrated **iOS** framework CallKit and **Android** ConnectionService and coded with **Objective-C** in **Xcode** and **Java** in Android Studio to realize a video call notification feature to increase match pick-up rate
- Performed local camera and microphone selection and checking features through Agora to ensure WebRTC video quality

Huawei

Shanghai, China

Algorithm Engineer Intern

July 2020 – September 2020

- Applied calibration and distortion correction algorithm to process video flows of camera sensor for self-driving vehicles
 - Achieved reduction of distortion correction errors from 1.395 ± 0.136 pixels to 0.055 ± 0.192 pixels through cylindrical projection and single-point mapping in **Python**
 - Increased processing efficiency of DSP by developing mapping table and video processing algorithms in **C** for digital signal processors based on **OpenCV**
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PROJECTS

Santorini Game

September 2022 - November 2022

- Crafted a strategy-based board game backend playable with 2 players by using **Java** and **Apache Maven**
- Designed the game system and applied the **MVC** design pattern for a more extensible and robust system
- Developed user interface via **React** and **TypeScript** and created **Restful API** for server-client data exchange
- Implemented unit tests and integration tests with over 90% test coverage using **JUnit** and **ts-jest**

Rubik's Cube Helper

November 2021 - November 2022

- Built a cross-platform app via **React**, **JavaScript** on web and **Swift** on **iOS** to make a more personalized Cube experience
- Implemented responsive design, accessibility development, and multiple interactions to meet WCAG 2.1 standard on web
- Delivered **Node.js** backend with Express, deployed on **AWS** to enable data persistence and provided real-time world records

EEG-based BCI Research (Digital Image Lab, Tongji University)

September 2019 – June 2021

- Constructed classification models of brain during working memory maintenance based on 128-channel EEG dataset for material-specific Sternberg task through feature selection and machine learning with Python
- Extracted features of EEG signals by empirical mode decomposition (EMD) and phase space reconstruction (PSR) in Matlab and executed SVM, KNN, Random Forest (RF), and nested cross-validation in Python
- Improved classification performance from 71.45% to 91.82% under SVM and published results in IEEE paper as first author

Course Selection Management System

June 2020 - June 2020

- Programmed a full-stack course selection management system with the separation of frontend and backend
 - Developed web app using **Vue.js** and Element-UI, and utilized **JavaScript**, **Node.js** and **MySQL** database for backend
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ACADEMIC PUBLICATION

Classification of EEG signals during Working-Memory Maintenance based on Phase Space Reconstruction of Empirical Mode Decomposition, Oct 2020, ISBN:978-1-6654-2299-4